INCH-POUND

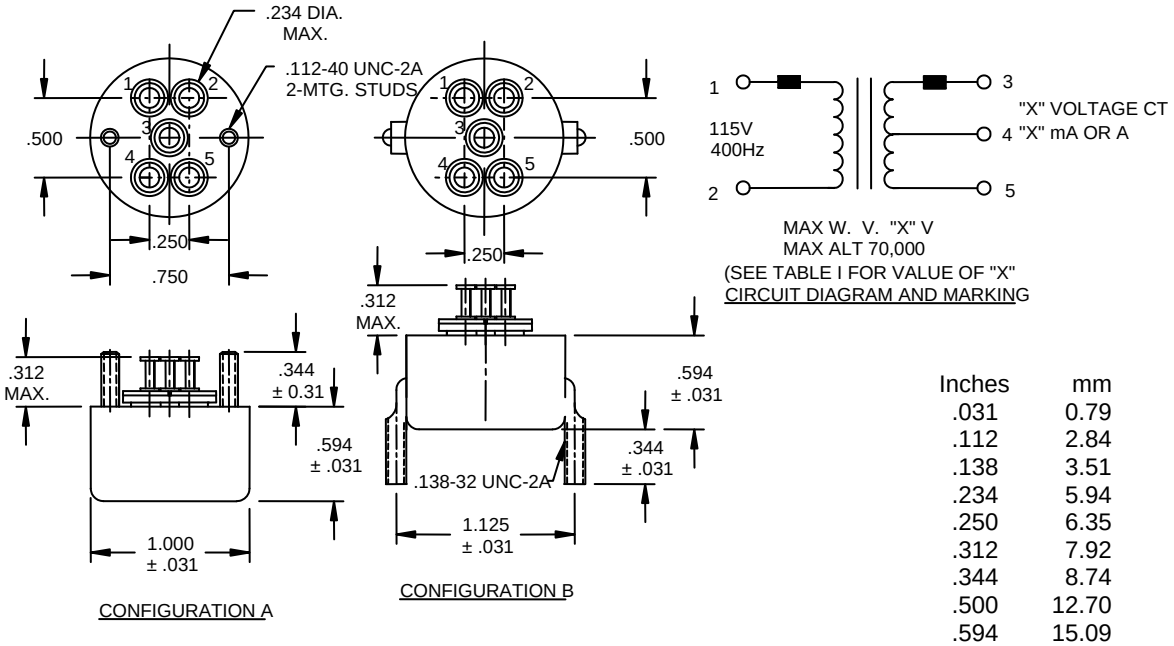
MIL-PRF-27/50D
29 September 2009
SUPERSEDING
MIL-PRF-27/50C
7 February 1984

PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 3 VOLT AMPERES, 400 HZ

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.



Inches	mm
.031	0.79
.112	2.84
.138	3.51
.234	5.94
.250	6.35
.312	7.92
.344	8.74
.500	12.70
.594	15.09
.750	19.05
1.000	25.40
1.125	28.58

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.

REQUIREMENTS: (When numbers in parentheses such as (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.25 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .312 inch (7.92 mm), maximum.

Terminal strength: Method 211 of MIL-STD-202, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 3.0 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 75 percent minimum at rated load.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 18 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 20°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 110°C.

Vibration: MIL-STD-202, method 204, test condition D.

Marking location: See figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.

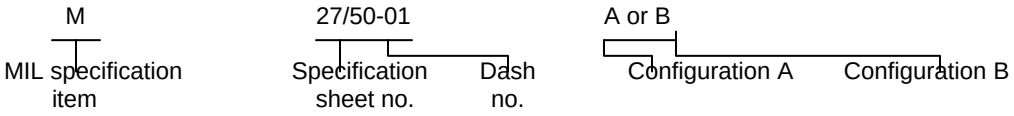
VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/50-44 shall be sufficient to grant qualification approval to M27/39-01 through M27/50-44.

Part or Identifying Number (PIN), example:



Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01 A or B	6	0.500	325
02 A or B	6.3	0.476	325
03 A or B	7	0.429	325
04 A or B	7.5	0.400	325
05 A or B	8	0.375	325
06 A or B	9	0.333	325
07 A or B	10	0.300	325
08 A or B	11	0.273	325
09 A or B	12	0.250	325
10 A or B	12.6	0.238	325
11 A or B	13	0.231	325
12 A or B	15	0.200	325
13 A or B	16	0.188	325
14 A or B	18	0.167	325
15 A or B	20	0.150	325
16 A or B	22	0.136	325
17 A or B	24	0.125	325
18 A or B	26	0.115	325
19 A or B	28	0.107	325
20 A or B	30	0.100	325
21 A or B	33	0.091	325
22 A or B	36	0.083	325
23 A or B	39	0.077	325
24 A or B	43	0.070	325
25 A or B	47	0.064	325
26 A or B	51	0.059	325
27 A or B	56	0.054	325
28 A or B	62	0.048	325
29 A or B	68	0.044	325
30 A or B	75	0.040	325
31 A or B	82	0.037	325
32 A or B	91	0.033	325
33 A or B	100	0.030	325
34 A or B	110	0.027	325
35 A or B	115	0.026	325
36 A or B	120	0.025	325
37 A or B	130	0.023	325
38 A or B	150	0.020	325
39 A or B	180	0.017	325
40 A or B	200	0.015	440
41 A or B	220	0.014	440
42 A or B	240	0.013	440
43 A or B	270	0.011	580
44 A or B	300	0.010	580

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-PRF-27/48	MIL-PRF-27/49	MIL-PRF-27/51	MIL-PRF-27/52
MIL-PRF-27/53	MIL-PRF-27/54	MIL-PRF-27/55	MIL-PRF-27/56
MIL-PRF-27/57	MIL-PRF-27/58	MIL-PRF-27/59	MIL-PRF-27/60
MIL-PRF-27/61	MIL-PRF-27/62	MIL-PRF-27/63	MIL-PRF-27/64
MIL-PRF-27/65	MIL-PRF-27/66	MIL-STD-202	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous Issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5950-2009-029)

Review activities:
Army - AR, CR4
Navy - AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T03:19Z
Check the source to verify that this is the current version before use.